

BAULKHAM HILLS HIGH SCHOOL

HALF YEARLY EXAMINATION

2009

YEAR 11

MATHEMATICS ADVANCED

Time Allowed:

Two hours
(plus 5 mins reading time)

Instructions:

- Write in the answer booklets provided
- Do not write on this question paper
- Show all working
- Use black or blue pens only
- Start a new page for each question
- Write your name and your teacher's name at the top of each page
- Board approved calculators may be used

Question 1 (10 marks)

- a) The retail price of a computer is \$940.50. This includes 10% GST.
What is the GST on the computer? 1
- b) Evaluate $\frac{2(1 - (\frac{2}{3})^7)}{1 - \frac{2}{3}}$ correct to 2 decimal places 2
- c) If $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$
- i) Find $1^3 + 2^3 + \dots + 10^3$ 1
- ii) Find $11^3 + 12^3 + \dots + 30^3$ 2
- d) Express $0.\dot{1}\dot{7}$ as a fraction in its simplest form. 2
- e) Light travels at 300000 km/s. How far does light travel in 1 year?
Give your answer in scientific notation, correct to 2 significant figures. 2

Question 2 (12 Marks)

- a) Simplify $\sqrt{8} - 2\sqrt{18}$ 1
- b) If $\sqrt{A} = \frac{3\sqrt{6}}{\sqrt{3}}$ find the value of A 2
- c) If $\frac{3\sqrt{2}-1}{\sqrt{2}+1} = a+b\sqrt{2}$ find the values of a and b 2
- d) Expand $(x-2)^3$ 2
- e) Factorise
- i) $x^2 - 5x - 6$ 1
- ii) $4x^2 - 100$ 2
- iii) $x^4 - x^3 - xy^3 + y^3$ 2

Question 3 (11 Marks)

a) Simplify:

i) $\frac{x}{x^2 - 2xy}$

1

ii) $\frac{1}{a+2} + \frac{1}{a^2-4}$

2

b) Solve:

i) $1 + \frac{6x}{5} = 4x$

2

ii) $\frac{2x+3}{2} - \frac{x+4}{4} = -3$

3

iii) $(1-2x)^2 = 9$

3

Question 4 (13 marks)

a) If $\cos \theta = \frac{2}{7}$ and $\sin \theta < 0$, find the exact value of $\tan \theta$

3

b) Find the exact value of $\sec 570^\circ$

2

c) Sketch the graph of the following, showing all essential features.

i) $y = 2^x + 1$

2

ii) $(x-2)^2 + (y+1)^2 = 4$

2

d) State the domain and range for each of the graphs in Question 4 c).

4

Question 5 (10 marks)

a) If $f(x) = \begin{cases} x^2 & x \leq 0 \\ \frac{x}{2} & 0 < x \leq 1 \\ \frac{1}{x} & x > 1 \end{cases}$

i) find $f(-2) + f(1) + f(1\frac{1}{2})$

2

ii) find $f(a^2 + 2)$

1

b) Solve

i) $9 - x^2 \leq 4$

2

ii) $|x - 2| = 2x + 3$

3

iii) $|4 - x| \leq 2$

2

Question 6 (10 marks)

a) Solve $\tan^2 \theta - 1 = 0$ for $0 \leq \theta \leq 360^\circ$

3

b) Prove

$$\frac{\cos \alpha}{1 + \sin \alpha} + \frac{\cos \alpha}{1 - \sin \alpha} = 2 \sec \alpha$$

2

c) If $\frac{6^{2n-1}}{18^{1-4n}} = 2^a \times 3^b$ find expressions for a and b in terms of n

3

d) Draw a function which has a domain of $x \neq 2$ and a range of $y < 0$ where x and y are real numbers

2

Question 7 (11 marks)

a) Solve simultaneously

$$y = \frac{10}{x+1} \quad \text{and} \quad y = x - 2$$

2

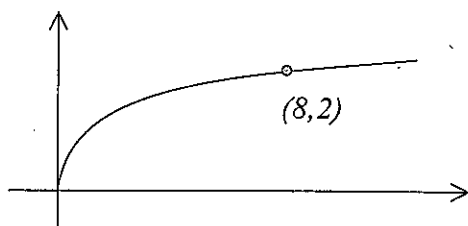
b) Simplify

$$\frac{a - a^{-1}}{1 - \frac{1}{a^{-2}}}$$

3

c) A person walks 10km west then 6km north.
What is his bearing **back to** his starting point?

3

d) Below is a part of a function $y = f(x)$ which is odd.

i) Copy and complete the graph.

1

ii) If $f(x) + 2 = 0$, find x

2

Question 8 (10 marks)

- a) Sketch the region which satisfies both inequalities.

$$xy \geq 1 \text{ and } y \leq \sqrt{4-x^2}$$

3

- b) i) Sketch on the same set of axes

$$y = 4 - x^2 \text{ and } y = 2|x - 2|$$

3

- ii) Hence, solve $4 - x^2 > 2|x - 2|$

1

- c) The parabola $y = ax^2 + bx + c$ has a vertex at (2,1) and passes through (0,0). Find a, b and c.

3

Question 9 (12 marks)

- a) If $f(x) = x^2$ and $g(x) = \frac{2}{x+1}$

- i) Show that if $g[f(x)] = f[g(x)]$ then $2x^2 - 4x + 2 = 0$

2

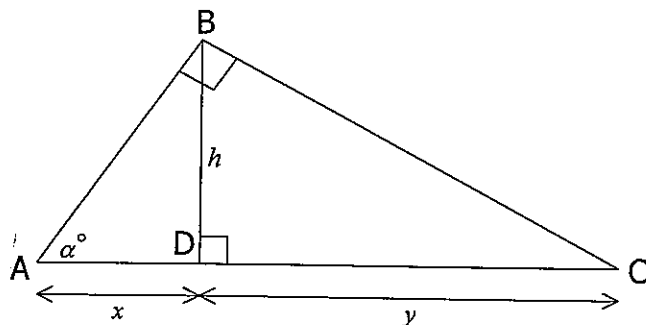
- ii) Hence find the values of x for which $g[f(x)] = f[g(x)]$

2

- b) If $x + y = 1$ and $x^3 + y^3 = 19$, find the value of $x^2 + y^2$

3

- c)



- i) By finding 2 expressions for h , show that $x \tan \alpha = y \tan(90 - \alpha)$

2

- ii) Hence, show $\frac{y}{x} = \tan^2 \alpha$

1

- iii) For what values of α is $y > x$?

2

Question 1: - [10]

a) $110\% = \$940.50$

$10\% = 940.50 \div 11$

$= \$85.50$ ①

b) $5.6488 \dots = 5.65$ ①

c) (i) $1^3 + \dots + 10^3 = \frac{10^2 (11)^2}{4}$
 $= 3025$ ①

(ii) $11^3 + \dots + 30^3 = S_{30} - S_{10}$
 $= \frac{(30)^2 (31)^2}{4} - 3025$
 $= 216225 - 3025$
 $= 213200$ ①

d) let $x = 0.1777 \dots$
 $10x = 1.777 \dots$
 $100x = 17.777 \dots$ ①
 $\therefore 90x = 16$
 $x = \frac{16}{90}$
 If $\frac{17}{99}$ ①
 $= \frac{8}{45}$ ①

Solutions: - Yr 11 2 Unit + 1/2 Yearly.

e) Dist. = $300000 \times 3600 \times 24 \times 365$ ①
 $= 9.5 \times 10^{12}$ ①

Question 2. [12]

a) $\sqrt{8} - 2\sqrt{18} = 2\sqrt{2} - 6\sqrt{2}$
 $= -4\sqrt{2}$ ①

b) $\sqrt{A} = \frac{3\sqrt{6}}{\sqrt{3}}$
 $= 3\sqrt{2}$
 $= \sqrt{18}$ ①
 $\therefore A = 18$ ①

c) $\frac{(3\sqrt{2}-1)(\sqrt{2}-1)}{(\sqrt{2}+1)(\sqrt{2}-1)}$ ①
 $= \frac{6 - 3\sqrt{2} - \sqrt{2} + 1}{2-1}$
 $= 7 - 4\sqrt{2}$
 $\therefore a = 7 \quad b = 4$ ①

d) $(x-2)^3 = (x-2)(x^2-4x+4)$
 1 off each error
 $= x^3 - 4x^2 + 4x - 2x^2 + 8x - 8$
 $= x^3 - 6x^2 + 12x - 8$ ②

e) (i) $x^2 - 5x - 6 = (x-6)(x+1)$
 (ii) $4x^2 - 100 = 4(x^2 - 25)$
 2 for 3 factors = $4(x+5)(x-5)$
 1 for 2 factors
 (iii) $x^4 - x^3 - xy^3 + y^3$
 $= x^3(x-1) - y^3(x-1)$
 $= (x^3 - y^3)(x-1)$ ①
 $= (x-y)(x^2 + xy + y^2)(x-1)$ ①

Question 3. [11]

a) (i) $\frac{x}{x^2 - 2xy} = \frac{x}{x(x-2y)}$
 $= \frac{1}{x-2y}$ ①

(ii) $\frac{1}{a+2} + \frac{1}{a^2-4}$
 $= \frac{a-2}{(a+2)(a-2)} + \frac{1}{(a+2)(a-2)}$ ①
 $= \frac{a-1}{(a+2)(a-2)}$ ①

$$3b) (i) 1 + \frac{6x}{5} = 4x$$

$$5 + 6x = 20x \quad (1)$$

$$14x = 5$$

$$x = \frac{5}{14} \quad (1)$$

$$(ii) \frac{2x+3}{2} - \frac{x+4}{4} = -3$$

$$\frac{4x+6 - x-4}{4} = -3 \quad (1)$$

$$3x + 2 = -12 \quad (1)$$

$$3x = -14$$

$$x = -\frac{14}{3} \quad (1)$$

$$(iii) (1-2x)^2 = 9$$

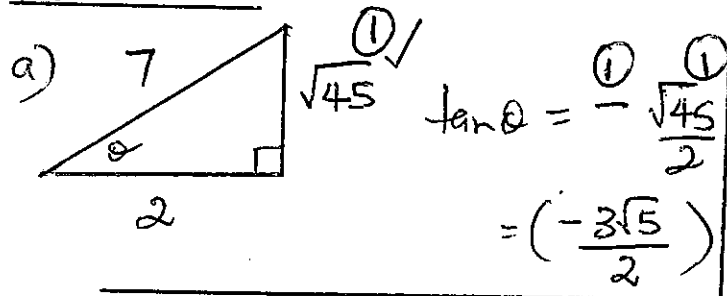
$$1-2x = \pm 3 \quad (1)$$

$$-2x = -1 \pm 3$$

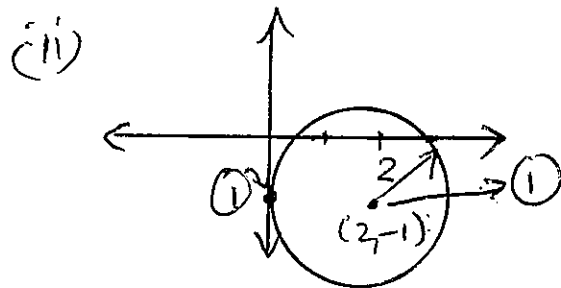
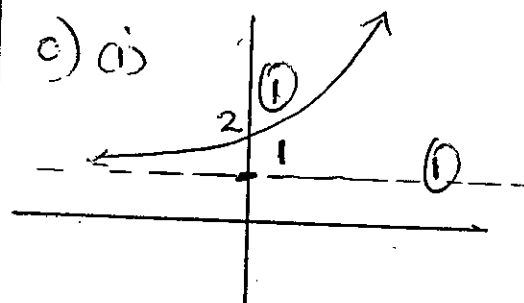
$$x = \frac{-1 \pm 3}{-2}$$

$$x = -1 \quad (1) \quad x = 2 \quad (1)$$

Question 4. [15]



b) $\sec 570^\circ = \frac{1}{\cos 210^\circ}$
 $= \frac{1}{-\cos 30^\circ}$
 $= \frac{1}{-\frac{\sqrt{3}}{2}} = -\frac{2}{\sqrt{3}} \quad (1)$



d) (i) D: all real x (1)
R: $y > 1$, y is real (1)

(ii) D: $0 \leq x \leq 4$, x is real (1)
R: $-3 \leq y \leq 1$, y is real (1)

Question 5. [10]

a) $f(-2) + f(1) + f(1\frac{1}{2})$
 $= (-2)^2 + \frac{1}{2} + \frac{1}{(\frac{3}{2})}$ For each error
 $= 5\frac{1}{6} \quad (2)$

(ii) $f(a^2+2) = \frac{1}{a^2+2} \quad (1)$

b) (i) $9 - x^2 \leq 4$
 $-x^2 \leq -5$
 $x^2 \geq 5$ -1 for wrong sign
 $x \leq -\sqrt{5}, x \geq \sqrt{5} \quad (2)$

(ii) $|x-2| = 2x+3$ -1 not checked
 $x-2 = 2x+3 \quad -x+2 = 2x+3$
 $-5 = 3x \quad -1 = 3x$
 $x = -\frac{5}{3} \quad x = -\frac{1}{3}$
Check $|-7| = -7$ False $|-2\frac{1}{3}| = 2\frac{1}{3}$ ✓
 $x = -\frac{1}{3} \quad (3)$

$$i) |4-x| \leq 2 \quad \text{1 off each error.}$$

$$\begin{aligned} -4+x &\leq 2 & 4-x &\leq 2 \\ x &\leq 6 & -x &\leq -2 \end{aligned}$$

$$x \geq 2$$

$$\therefore 2 \leq x \leq 6 \quad (2)$$

Question 6. 10 45, 225. 1

$$i) \tan^2 \theta - 1 = 0 \quad \text{1 off each error.}$$

$$\tan \theta = \pm 1$$

$$\theta = 45, 135, 225, 315. \quad (3)$$

$$i) \frac{\cos x}{1 + \sin x} + \frac{\cos x}{1 - \sin x} = 2 \sec x$$

LHS =

$$\frac{\cos x - \sin x \cos x + \cos x + \sin x \cos x}{1 - \sin^2 x} \quad (1)$$

$$1 - \sin^2 x$$

$$= \frac{2 \cos x}{\cos^2 x} \quad (1)$$

$$= 2 \sec x$$

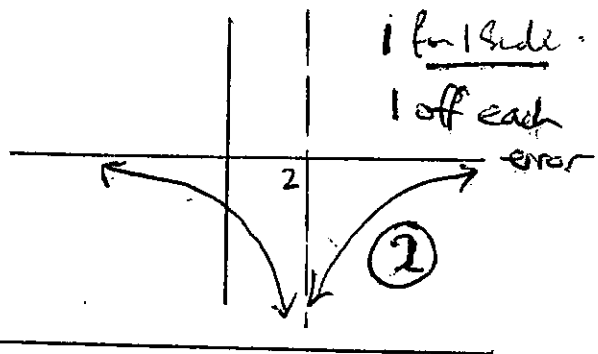
$$= \text{RHS.}$$

$$i) \frac{6^{2n-1}}{18^{1-4n}} = \frac{2^{2n-1} \cdot 3^{2n-1}}{2^{1-4n} \cdot (3^2)^{1-4n}} \quad (1)$$

$$= 2^{2n-1-(1-4n)} \cdot 3^{2n-1-(2-8n)}$$

$$= 2^{6n-2} \cdot 3^{10n-3} \quad (1)$$

d)



Question 7. 11

$$a) y = \frac{10}{x+1} \quad b) y = x-2$$

$$x-2 = \frac{10}{2x+1}$$

$$x^2 - 2x + x - 2 = 10$$

$$x^2 - x - 12 = 0$$

$$(x-4)(x+3) = 0$$

$$x = 4, -3 \quad (1)$$

$$y = 2, -5 \quad (1)$$

$$(4, 2), (-3, -5)$$

$$b) \frac{a - a^{-1}}{1 - \frac{1}{a^{-2}}} = \frac{a - \frac{1}{a}}{1 - a^2}$$

$$= \frac{a^2 - 1}{1 - a^2}$$

$$\text{1 off each error} \quad \frac{a}{1 - a^2}$$

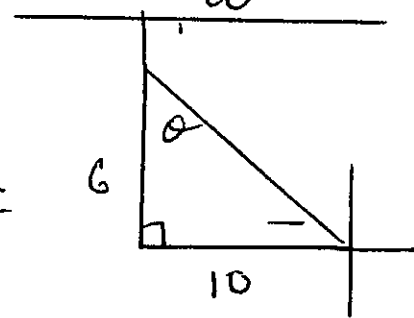
$$2 \text{ for } \frac{a-1}{a(1-a)} = \frac{(a+1)(a-1)}{a(1-a)}^{-1}$$

$$2 \text{ for answer } a \frac{(1-a)(1+a)}{1-a^2}$$

$$= -\frac{1}{a} \quad (3)$$

c)

1 off each error



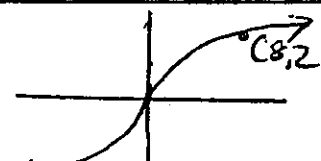
$$\tan \theta = \frac{10}{6}$$

$$\theta = 59^\circ 2' \quad (1)$$

$$\therefore \text{Bearing} = 180 - 59^\circ 2' \quad (1)$$

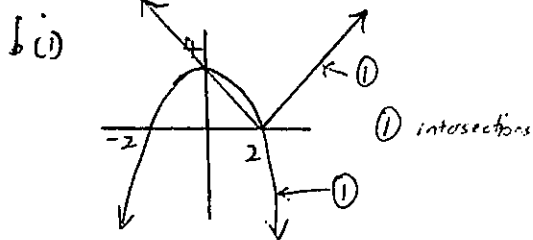
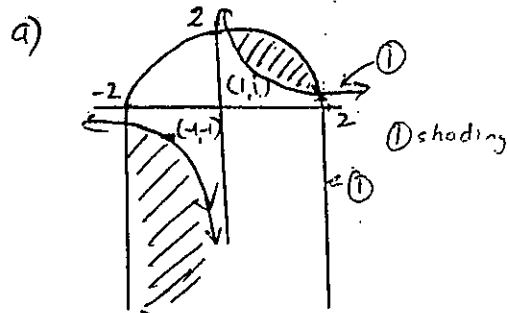
$$= 120^\circ 58' \quad (1)$$

d) (i)

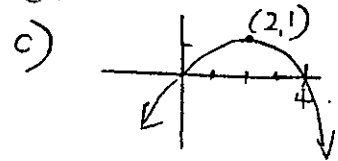


(diii) If $f(x)+2=0$
 $f(x) = -2$ ①
 Since it's odd $f(-8) = -2$.
 $\therefore x = -8$ ①

Question 8 10 $xy \geq 1$ $y \leq \sqrt{4-x^2}$



ai) $0 < x < 2$ ①



$y = ax + bx + c$
 $(0,0)$ satisfies $\Rightarrow c = 0$ ①
 $(2,1)$ " $\therefore 1 = 4a + 2b$ ②
 also $(4,0)$ satisfies

$\therefore$ ① $\begin{cases} 0 = 16a + 4b \\ 2 = 8a + 4b \end{cases}$ ②
 ② - ③ $-2 = 8a$
 $a = -\frac{1}{4}$
 $b = 1$
 $\therefore a = -\frac{1}{4}$ $b = 1$ $c = 0$ ①

Question 9. $f(x) = x^2$

$g(x) = \frac{2}{x+1}$
 $g[f(x)] = g(x^2)$
 $= \frac{2}{x^2+1}$
 $f[g(x)] = f\left(\frac{2}{x+1}\right)$
 $= \left(\frac{2}{x+1}\right)^2$
 $= \frac{4}{(x+1)^2}$
 $\frac{2}{x^2+1} = \frac{4}{(x+1)^2}$ ① for both

$2(x+1)^2 = 4(x^2+1)$
 $2x^2 + 4x + 2 = 4x^2 + 4$ ①
 $2x^2 - 4x + 2 = 0$
 $2(x-1)^2 = 0$
 $\therefore x = 1$.

b) $x+y=1$ $x^3+y^3=19$
 $x^3+y^3 = (x+y)(x^2-xy+y^2)$
 $19 = 1(x^2-xy+y^2)$
 $\therefore x^2-xy+y^2 = 19$ ①
 now $(x+y)^2 = 1$
 $\therefore x^2+2xy+y^2 = 1$ ②
 ① - ② $3xy = -18$
 $xy = -6$ ③

sub. into ①
 $x^2+6+y^2 = 19$
 $x^2+y^2 = 13$ ④

90 (i) $\tan \alpha = \frac{h}{x}$
 $\therefore h = x \tan \alpha$ ①

$\angle BCD = 90 - \alpha$

$\therefore \tan(90 - \alpha) = \frac{h}{y}$

$h = y \tan(90 - \alpha)$ ①

(ii) $x \tan \alpha = y \tan(90 - \alpha)$

$x \tan \alpha = y \cot \alpha$

$x \tan \alpha = \frac{y}{\tan \alpha}$ ①

$\frac{y}{x} = \tan^2 \alpha$

(iii) $y > x$

then $\frac{y}{x} > 1$

$\therefore \tan^2 \alpha > 1$ ①

α is acute
 $\therefore 45 < \alpha < 90$ ①